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SRN: PES2UG24CS311
Week: 7
Class: F

1. ALU:

a) Test-Bench File:

```
alu_tb.v
alu_tb.v
1 `timescale 1 ns / 1 ps
2 `define TESTVECS 16
3 module tb;
4 reg clk, reset;
5 reg [1:0] op;
6 reg [15:0] i0, i1;
7 wire [15:0] o;
8 wire cout;
9 reg [33:0]
10 test_vecs [0:(`TESTVECS-1)];
11 integer i;
12 initial
13 begin
14 $dumpfile("alu.vcd");
15 $dumpvars(0,tb);
16 end
17 initial
18 begin
19 reset = 1'b1;
20 #12.5
21 reset = 1'b0;
22 end
23 initial clk = 1'b0;
24 always
25 #5
26 clk =~ clk;
27 initial
28 begin
29 test_vecs[0][33:32] = 2'b00; test_vecs[0][31:16] = 16'h0000; test_vecs[0][15:0] = 16'h0000;
30 test_vecs[1][33:32] = 2'b00; test_vecs[1][31:16] = 16'haa55; test_vecs[1][15:0] = 16'h55aa;
31 test_vecs[2][33:32] = 2'b00; test_vecs[2][31:16] = 16'hffff; test_vecs[2][15:0] = 16'h0001;
32 test_vecs[3][33:32] = 2'b00; test_vecs[3][31:16] = 16'h0001; test_vecs[3][15:0] = 16'h7fff;
33
34 test_vecs[4][33:32] = 2'b01; test_vecs[4][31:16] = 16'h0000; test_vecs[4][15:0] = 16'h0000;
35 test_vecs[5][33:32] = 2'b01; test_vecs[5][31:16] = 16'haa55; test_vecs[5][15:0] = 16'h55aa;
36 test_vecs[6][33:32] = 2'b01; test_vecs[6][31:16] = 16'hffff; test_vecs[6][15:0] = 16'h0001;
37 test_vecs[7][33:32] = 2'b01; test_vecs[7][31:16] = 16'h0001; test_vecs[7][15:0] = 16'h7fff;
38 test_vecs[8][33:32] = 2'b10; test_vecs[8][31:16] = 16'h0000; test_vecs[8][15:0] = 16'h0000;
39 test_vecs[9][33:32] = 2'b10; test_vecs[9][31:16] = 16'haa55; test_vecs[9][15:0] = 16'h55aa;
40 test_vecs[10][33:32] = 2'b10; test_vecs[10][31:16] = 16'hffff; test_vecs[10][15:0] = 16'h0001;
41 test_vecs[11][33:32] = 2'b10; test_vecs[11][31:16] = 16'h0001; test_vecs[11][15:0] = 16'h7fff;
42 test_vecs[12][33:32] = 2'b11; test_vecs[12][31:16] = 16'h0000; test_vecs[12][15:0] = 16'h0000;
43 test_vecs[13][33:32] = 2'b11; test_vecs[13][31:16] = 16'haa55; test_vecs[13][15:0] = 16'h55aa;
44 test_vecs[14][33:32] = 2'b11; test_vecs[14][31:16] = 16'hffff; test_vecs[14][15:0] = 16'h0001;
45 test_vecs[15][33:32] = 2'b11; test_vecs[15][31:16] = 16'h0001; test_vecs[15][15:0] = 16'h7fff;
46
47 end
48 initial {op, i0, i1} = 0;
49 alu alu_0 (op, i0, i1, o, cout);
50 initial begin
51 #6
52 for (i = 0; i < `TESTVECS; i = i + 1) begin
53 #10 {op, i0, i1} = test_vecs[i];
54 $display("TestVec: %2d | OP: %b | Input1: %h | Input2: %h | Output: %h |", i, op, i0, i1, o);
55 end
56 #50 $finish;
57 end
58 endmodule
```

```
File Edit Selection View Go Run ... LAB-VII
alu_tb.v
28 begin
29 test_vecs[0][33:32] = 2'b00; test_vecs[0][31:16] = 16'h0000; test_vecs[0][15:0] = 16'h0000;
30 test_vecs[1][33:32] = 2'b00; test_vecs[1][31:16] = 16'haa55; test_vecs[1][15:0] = 16'h55aa;
31 test_vecs[2][33:32] = 2'b00; test_vecs[2][31:16] = 16'hffff; test_vecs[2][15:0] = 16'h0001;
32 test_vecs[3][33:32] = 2'b00; test_vecs[3][31:16] = 16'h0001; test_vecs[3][15:0] = 16'h7fff;
33 test_vecs[4][33:32] = 2'b01; test_vecs[4][31:16] = 16'h0000; test_vecs[4][15:0] = 16'h0000;
34 test_vecs[5][33:32] = 2'b01; test_vecs[5][31:16] = 16'haa55; test_vecs[5][15:0] = 16'h55aa;
35 test_vecs[6][33:32] = 2'b01; test_vecs[6][31:16] = 16'hffff; test_vecs[6][15:0] = 16'h0001;
36 test_vecs[7][33:32] = 2'b01; test_vecs[7][31:16] = 16'h0001; test_vecs[7][15:0] = 16'h7fff;
37 test_vecs[8][33:32] = 2'b10; test_vecs[8][31:16] = 16'h0000; test_vecs[8][15:0] = 16'h0000;
38 test_vecs[9][33:32] = 2'b10; test_vecs[9][31:16] = 16'haa55; test_vecs[9][15:0] = 16'h55aa;
39 test_vecs[10][33:32] = 2'b10; test_vecs[10][31:16] = 16'hffff; test_vecs[10][15:0] = 16'h0001;
40 test_vecs[11][33:32] = 2'b10; test_vecs[11][31:16] = 16'h0001; test_vecs[11][15:0] = 16'h7fff;
41 test_vecs[12][33:32] = 2'b11; test_vecs[12][31:16] = 16'h0000; test_vecs[12][15:0] = 16'h0000;
42 test_vecs[13][33:32] = 2'b11; test_vecs[13][31:16] = 16'haa55; test_vecs[13][15:0] = 16'h55aa;
43 test_vecs[14][33:32] = 2'b11; test_vecs[14][31:16] = 16'hffff; test_vecs[14][15:0] = 16'h0001;
44 test_vecs[15][33:32] = 2'b11; test_vecs[15][31:16] = 16'h0001; test_vecs[15][15:0] = 16'h7fff;
45
46 end
47 initial {op, i0, i1} = 0;
48 alu alu_0 (op, i0, i1, o, cout);
49 initial begin
50 #6
51 for (i = 0; i < `TESTVECS; i = i + 1) begin
52 #10 {op, i0, i1} = test_vecs[i];
53 $display("TestVec: %2d | OP: %b | Input1: %h | Input2: %h | Output: %h |", i, op, i0, i1, o);
54 end
55 #50 $finish;
56 end
57 endmodule
```

b)VVP

```
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/P...
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-VII$ iverilog lib.v alu.v alu_tb.v -o alu
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-VII$ vvp alu
VCD info: dumpfile alu.vcd opened for output.
|TestVec: 0 | OP: 00 | Input1: 0000 | Input2: 0000 | Output: 0000 |
|TestVec: 1 | OP: 00 | Input1: aa55 | Input2: 55aa | Output: 0000 |
|TestVec: 2 | OP: 00 | Input1: ffff | Input2: 0001 | Output: ffff |
|TestVec: 3 | OP: 00 | Input1: 0001 | Input2: 7fff | Output: 0000 |
|TestVec: 4 | OP: 01 | Input1: 0000 | Input2: 0000 | Output: 8000 |
|TestVec: 5 | OP: 01 | Input1: aa55 | Input2: 55aa | Output: 0000 |
|TestVec: 6 | OP: 01 | Input1: ffff | Input2: 0001 | Output: 54ab |
|TestVec: 7 | OP: 01 | Input1: 0001 | Input2: 7fff | Output: fffe |
|TestVec: 8 | OP: 10 | Input1: 0000 | Input2: 0000 | Output: 8002 |
|TestVec: 9 | OP: 10 | Input1: aa55 | Input2: 55aa | Output: 0000 |
|TestVec: 10 | OP: 10 | Input1: ffff | Input2: 0001 | Output: 0000 |
|TestVec: 11 | OP: 10 | Input1: 0001 | Input2: 7fff | Output: 0001 |
|TestVec: 12 | OP: 11 | Input1: 0000 | Input2: 0000 | Output: 0001 |
|TestVec: 13 | OP: 11 | Input1: aa55 | Input2: 55aa | Output: 0000 |
|TestVec: 14 | OP: 11 | Input1: ffff | Input2: 0001 | Output: ffff |
|TestVec: 15 | OP: 11 | Input1: 0001 | Input2: 7fff | Output: ffff |
alu_tb.v:54: $finish called at 216000 (1ps)
neeraj-r-rugi@neeraj-ubuntu: /media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-VII$
```

C)GtkWave:



